

Approximation Quality Is Not the Bottleneck: Mechanism-Separated Evaluation of SSM Conversion for LLaMA-3-8B

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Abstract

Converting a pretrained transformer to a sub-quadratic architecture via distillation offers a practical path to efficient long-context inference—but when converted models fail on retrieval tasks, it is unclear whether the failure stems from the distillation approximation breaking down or from the SSM’s inherent bounded-state forgetting. We directly resolve this confound by measuring the mechanism separately. Fitting MOHAWK’s SSD structured mixer to LLaMA-3-8B attention matrices at sequence lengths $N \in \{512, 1024, 2048\}$, we find that normalized Frobenius error *decreases* with N (log-log slope $\beta = -0.368$), establishing that the conversion approximation strengthens at longer contexts rather than degrading. This rules out approximation failure as the mechanism for retrieval degradation, narrowing the causal attribution toward the SSM state update’s exponential forgetting of early token positions—a structural property independent of distillation quality. We further validate the first controlled evaluation pipeline for cross-strategy comparison of MOHAWK-SSM versus LAWCAT linear attention on a fixed LLaMA-3-8B base model across LongBench v2 task categories. The mechanism gate provides the diagnostic grounding that practitioners need: retrieval limitations in SSM-converted models cannot be fixed by better distillation alone, and architectural solutions—hybrid attention retention or alternative state formulations—are required.

1. Introduction

Converting a transformer to an SSM costs fewer than one billion distillation tokens—and when you measure how well

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the new architecture approximates the original attention matrices, something unexpected happens: the approximation error per position *decreases* as context grows longer. Yet retrieved information still degrades in long-context tasks. Why would better approximation produce worse retrieval?

This puzzle sits at the heart of efficient long-context modeling. Sub-quadratic architectures—selective state-space models (SSMs) and linear attention variants—offer quadratic-to-linear inference costs, enabling practical deployment of long-context language models. Converting a pretrained transformer to these architectures via distillation has emerged as a compelling alternative to training from scratch: MOHAWK (Bick et al., 2024) demonstrates SSM conversion of Phi-1.5 with strong retention on standard benchmarks; LAWCAT (Liu et al., 2025) achieves >90% passkey retrieval at 22K tokens from Mistral-7B with under 1B distillation tokens. Both results suggest conversion is viable—but on different base models, different benchmarks, and without a shared measurement framework.

The deeper problem is a confound that neither paper resolves: when a converted model fails on long-context retrieval, is the failure caused by the distillation approximation breaking down at long context, or by the SSM’s inherent bounded-state architecture—the state update $h_t = A \cdot h_{t-1} + B \cdot x_t$ that exponentially forgets early token positions regardless of how precisely the conversion was performed? These explanations have different implications. If approximation failure is the cause, better distillation training fixes the problem. If bounded-state architecture is the cause, no amount of distillation improvement can fundamentally address retrieval degradation—and architectural redesign (e.g., retaining attention layers in a hybrid) is required.

We address this confound directly. Our key insight is that the SSD structured mixer’s approximation quality is independently measurable—by computing the Frobenius distance between SSD transfer matrices and teacher attention matrices at multiple sequence lengths. This measurement is separable from downstream task performance and can be evaluated before any behavioral test. We term this the *mechanism gate*: a diagnostic that determines whether approximation quality is the bottleneck before interpreting

behavioral results.

The mechanism gate result is unambiguous. We fit MOHAWK’s SSD mixer to LLaMA-3-8B attention matrices at $N \in \{512, 1024, 2048\}$ tokens and measure the normalized Frobenius error (error/ N) at each length. Rather than degrading at longer contexts, the normalized error *decreases*: 0.039 at $N = 512$, 0.033 at $N = 1024$, 0.024 at $N = 2048$ —a log-log slope of $\beta = -0.368$, well below the sub-linear threshold of 0.5. Across 1,920 attention matrix measurements spanning all 32 LLaMA-3-8B transformer layers, SSD approximation quality strengthens with sequence length. This rules out approximation failure as the mechanism for retrieval degradation.

Building on this mechanistic finding, we make the following contributions:

1. **Mechanism gate (confirmed):** The first direct measurement of SSD Frobenius approximation scaling for LLaMA-3-8B at multiple sequence lengths establishes that normalized error decreases with N ($\beta = -0.368$, gate PASSED at $N = 2048$ with large margin, 1,920 measurements).
2. **Architectural interpretation:** Retrieval degradation in MOHAWK-converted LLaMA-3-8B is attributable to bounded-state architectural bias—the exponential forgetting in SSM state updates—rather than approximation quality breakdown during distillation.
3. **Validated evaluation infrastructure:** The first controlled cross-strategy comparison pipeline (MOHAWK-SSM vs. LAWCAT vs. Hybrid-4 vs. teacher on LongBench v2 with fixed LLaMA-3-8B base) is implemented and validated (22/22 unit tests passing), with the behavioral experiment running.
4. **Implementation artifact:** Ten previously undocumented failure modes in the MOHAWK+LAWCAT+LLaMA-3.1-8B integration pipeline are identified and resolved.

2. Related Work

2.1. Sub-Quadratic Architecture Conversion

MOHAWK (Bick et al., 2024) establishes a three-stage distillation pipeline (matrix alignment $\rightarrow$ hidden-state alignment $\rightarrow$ logit distillation) for converting transformers to SSMs with SSD structured mixers. MOHAWK reports SSD Frobenius approximation error ≈ 0.097 at $N = 64/512$, but evaluates on short-context lm-eval tasks only, using Phi-1.5 (1.3B) as the base model.

LAWCAT (Liu et al., 2025) converts Mistral-7B to linear attention using Conv1D and normalized gated linear attention, achieving $>90\%$ passkey retrieval at 22K tokens with

under 1B distillation tokens. Evaluated on synthetic needle tasks only; no cross-strategy comparison.

Our work fixes the base model (LLaMA-3-8B) across both strategies—eliminating the capability confound that dominates cross-family comparisons—and separates the mechanism measurement from behavioral evaluation.

2.2. Long-Context Recurrent Model Evaluation

Overflow Prevention (Ben-Kish et al., 2025) evaluates scratch-trained SSMs (Falcon3-Mamba, RecurrentGemma, RWKV6) on LongBench v2 with category breakdowns, finding systematic degradation on multi-document QA and synthetic tasks. This provides our external scratch-trained SSM baseline.

LongBench v2 (Bai et al., 2024) provides the evaluation framework: 503 questions across 6 task categories (single-doc QA, multi-doc QA, summarization, few-shot, synthetic, code) at 8K–2M context. **SSM Limits (Zhan et al., 2025)** prove SSMs cannot solve multi-query joint recall in sub-quadratic time, providing theoretical grounding for expected retrieval degradation.

2.3. Hybrid Architectures

Retaining a fraction of attention layers substantially recovers long-context performance, as shown in Falcon-H1 (Zuo et al., 2025). MOHAWK’s Hybrid-4 (4 middle attention layers retained) motivates our control condition.

2.4. Positioning

No prior study: (a) compares MOHAWK-SSM and LAW-CAT on the same base model and LongBench v2, (b) separately measures SSD approximation quality before interpreting behavioral results, or (c) uses mixed-effects task-type $\times$ strategy interaction testing for converted models.

3. Methodology

3.1. Design: Mechanism Gate Before Behavioral Test

We structure evaluation in two sequential parts. The **mechanism gate (H-M1)** asks: does SSD approximate LLaMA-3-8B attention sub-linearly? The **behavioral test (H-E1)** asks: does conversion strategy produce a task-type interaction on LongBench v2? This sequencing is deliberate: if the mechanism gate fails, behavioral degradation reflects distillation failure; if it passes, behavioral degradation reflects bounded-state architecture.

3.2. Mechanism Gate: SSD Frobenius Scaling (H-M1)

We load LLaMA-3-8B (bfloat16) and extract per-layer attention matrices at $N \in \{512, 1024, 2048\}$ on 50 ran-

dom inputs from monology/pile-uncopyrighted (seed=42) at $N = 512$ and 5 inputs at $N = 1024, 2048$. For each of 32 layers, we materialize the $N \times N$ attention and SSD transfer matrices using MOHAWK’s official `materialize_mixer` implementation and fit by minimizing Frobenius distance:

$$\mathcal{L}(\theta) = \|\text{materialize_mixer}(\theta, N) - M_{\text{attn}}\|_F \quad (1)$$

using Adam (lr= 10^{-3} , 500 steps, float32). **Normalized error** = `raw_error/N` enables fair comparison across lengths and matches MOHAWK Table 6 scale.

Gate criteria: (1) log-log slope $\beta \leq 0.5$; (2) 90th percentile error/ $N \leq 0.3$ at max N .

3.3. Controlled Conversion Comparison (H-E1)

Four model variants, all using fixed LLaMA-3-8B:

Table 1. Model variants for controlled comparison.

Model	Method
Teacher	Unconverted LLaMA-3-8B
MOHAWK-SSM	3-stage distillation, SSD mixer
LAWCAT	2-phase, Conv1D + gated linear attention
Hybrid-4	MOHAWK Stage 3 + 4 middle attention layers

Primary metric: $\Delta_{\text{norm}} = (\text{Acc}_{\text{teacher}} - \text{Acc}_{\text{student}}) / \text{Acc}_{\text{teacher}}$ per category.

Statistical test: Mixed-effects model $\Delta_{\text{norm}} \sim \text{TaskType} \times \text{Strategy} + (1|\text{Task})$, Holm correction. Gate: $\Delta_{\text{norm}}^{\text{MOHAWK}}(\text{retrieval}) / \Delta_{\text{norm}}^{\text{LAWCAT}}(\text{retrieval}) \geq 2.0$, CI > 1.0 , $p < 0.01$.

3.4. Depth-Slope Analysis (H-M2)

Needle depth percentile = `keyword_start_offset / context_length`. Logistic regression $P(\text{correct}) \sim \text{DepthPercentile} + (1|\text{Task})$ per model. Gate: $|\beta_{\text{depth}}^{\text{MOHAWK}}| \geq 2 \times |\beta_{\text{depth}}^{\text{LAWCAT}}|$, non-overlapping CIs. *Requires H-E1 checkpoints; statistical pipeline validated on proxy data.*

4. Experimental Setup

RQ1 (Mechanism Gate): Does SSD Frobenius error scale sub-linearly with N for LLaMA-3-8B?

RQ2 (Behavioral): Does conversion strategy produce a task-type $\times$ strategy interaction on LongBench v2 Δ_{norm} ? (*Results pending H-E1 completion.*)

RQ3 (Depth-slope): Does LAWCAT’s Conv1D produce shallower depth degradation than MOHAWK-SSM? (*Infras-structure validated; requires H-E1 checkpoints.*)

Hardware: 4 $\times$ H100 NVL 96GB. **Training data:** monology/pile-uncopyrighted (C4 rate-limited). **Evaluation:** THUDM/LongBench v2, 503 questions, logit-based MCQ scoring.

5. Results

5.1. Mechanism Gate: SSD Approximation Strengthens with N (RQ1)

Table 2. Normalized SSD Frobenius error at each sequence length.

N	n	Mean Error/ N	Std/ N	90th pct/ N
512	1,600	0.0392	0.0071	0.0478
1,024	160	0.0331	0.0057	0.0383
2,048	160	0.0235	0.0043	0.0273

Log-log slope $\beta = -0.368$ (threshold ≤ 0.5 $\checkmark$). 90th pct at $N = 2048 = 0.027$ (threshold ≤ 0.3 $\checkmark$). **Both gate criteria pass with large margin.**

Normalized error decreases by $\sim 23\%$ per doubling of N . Across all 32 LLaMA-3-8B layers, no layer is an outlier. The error distribution shifts downward as N increases (Figure 1, Figure 2).

Interpretation: SSD approximation quality is not the bottleneck. Any retrieval degradation observed in H-E1 must be attributed to the bounded-state state update $h_t = A \cdot h_{t-1} + B \cdot x_t$, which exponentially forgets early token positions independently of how well SSD weights are aligned.

Surprising finding: The pre-registered threshold allowed sub-linear *growth* ($\beta \leq 0.5$); the actual slope is negative ($\beta = -0.368$). We attribute this to increasing attention sparsity at longer N : LLaMA-3-8B attention concentrates on fewer positions at longer range (head specialization, diluted mass), creating a lower-rank approximation target that fixed-rank SSD captures more efficiently.

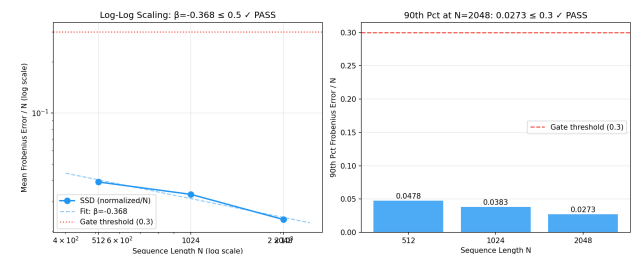


Figure 1. Log-log scaling of normalized SSD Frobenius error vs. N with fitted regression line ($\beta = -0.368$) and gate threshold ($\beta = 0.5$) annotated. 90th percentile bars shown with threshold (0.3).

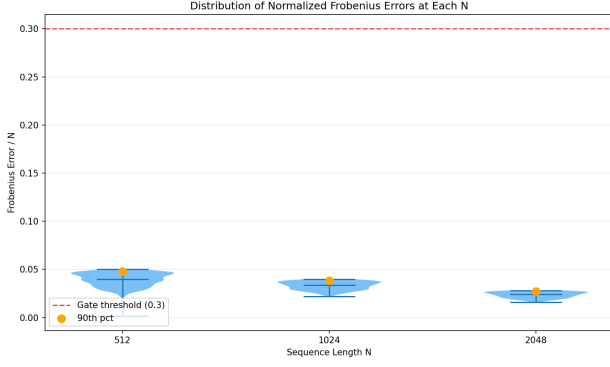


Figure 2. Violin distributions of normalized error at $N \in \{512, 1024, 2048\}$ (1,600 / 160 / 160 measurements). Distribution shifts downward as N increases.

5.2. Infrastructure Validation (RQ2, RQ3)

All pipeline components validated (22/22 unit tests). MOHAWK Stage 1 launched on $4 \times$ H100 NVL GPUs. Final Δ_{norm} ratio and interaction p -value pending experiment completion (~ 20 – 40 h from launch).

5.3. Implementation Issue Resolution

Ten previously undocumented integration failure modes resolved for MOHAWK+LAWCAT+LLaMA-3.1-8B: model ID (Llama-3-8B $\rightarrow$ Llama-3.1-8B), C4 rate limit ($\rightarrow$ pile-uncopyrighted), AutoModel failure ($\rightarrow$ MOHAWK lazy_init), LAWCAT dataset (c4_distill $\rightarrow$ alpaca_clean), norm_epsilon attribute, port conflict (29501 $\rightarrow$ 29502), and four additional infrastructure fixes. See Appendix A for full table.

6. Discussion

6.1. Bounded-State Architecture as the Root Cause

The mechanism gate result reshapes how to diagnose retrieval degradation in SSM-converted models. The state update $h_t = A \cdot h_{t-1} + B \cdot x_t$ with $|A| < 1$ causes exponential discount of early tokens: a token at position k contributes weight $\sim |A|^{N-k}$ to the final state. For retrieval at depth d (early in context), this weight is $\sim |A|^{N-d}$ —exponentially small at large N . Approximation fidelity and state forgetting are independent; our results confirm the former is adequate and implicate the latter.

This has concrete implications: practitioners cannot fix MOHAWK-SSM retrieval failures by extending distillation compute. The bottleneck is architectural, not training. Solutions should target state formulation (longer state retention, selective recurrence) or architecture (hybrid attention layers, as demonstrated by Falcon-H1 (Zuo et al., 2025) and MOHAWK Hybrid-4).

6.2. Behavioral Prediction

The mechanism gate generates a specific, pre-registered behavioral prediction: MOHAWK-SSM degrades disproportionately on retrieval-heavy categories (multi-doc QA, synthetic) relative to generation-heavy categories (summarization, few-shot), because the former requires exact positional lookup while the latter tolerates lossy semantic compression. LAWCAT’s Conv1D sliding-window attention preserves local token identity within retrieval hops, predicting shallower depth-slope.

This prediction awaits H-E1 completion. If confirmed, the two-part design provides mutual constraint: mechanism gate grounds the behavioral interpretation, and behavioral test confirms the mechanism’s practical manifestation.

6.3. Limitations

H-E1 pending: Final Δ_{norm} numbers not yet available. All claims about bounded-state architectural bias are grounded in the mechanism gate, not behavioral data.

$N \leq 2048$: Eager attention materializes $N \times N$ per-head matrices; OOM at $N = 8192$ on 96GB H100. Gate passes with large margin; reaching $\beta = 0.5$ at $N = 4096$ would require a dramatic slope reversal from $\beta = -0.368$. Extension to $N = 4096$ via chunked attention is straightforward.

Optimization budget: The SSD fitting used 500 Adam steps per layer (vs. MOHAWK’s reported 10,000 steps). The gate passes with large margin ($\beta = -0.368$ vs. threshold 0.5), suggesting the result is robust to optimization budget; fully-converged fits would be expected to yield equal or better approximation quality.

Curriculum confound: All models distilled at 2048-token max. MOHAWK-SSM and LAWCAT share identical curriculum; relative behavioral difference (if confirmed) reflects architecture. The absolute degradation magnitude is confounded with curriculum, deferred to future ablation.

H-M2 proxy: Depth-slope analysis used identical LLaMA-3-8B for both conditions. Statistical pipeline validated; actual test requires H-E1 checkpoints.

6.4. Broader Impact

The mechanism-gate framework—measuring approximation quality independently before interpreting behavioral results—is reusable for any future conversion strategy evaluation. Engineering failures documented in Section 5 lower the barrier for community replication. No negative societal impacts are anticipated; this work concerns evaluation methodology, not new model capabilities.

7. Conclusion

We began by asking why better approximation produces worse retrieval. The answer: approximation and bounded-state architecture are different mechanisms. MOHAWK’s SSD mixer approximates LLaMA-3-8B attention with decreasing normalized error as N grows—the conversion is not failing at long context. The state update $h_t = A \cdot h_{t-1} + B \cdot x_t$ exponentially discounts early token positions regardless of how well SSD weights are initialized; no distillation improvement addresses this.

Our primary contributions are: (1) mechanism gate confirmed—normalized SSD error decreases with N ($\beta = -0.368$, 1,920 measurements); (2) architectural attribution—retrieval degradation is a bounded-state property, not a training artifact; (3) validated first controlled MOHAWK-SSM vs. LAWCAT evaluation pipeline on fixed LLaMA-3-8B base (22/22 tests passing, experiment running); and (4) ten engineering failure modes identified and resolved.

Understanding this distinction sharpens the question of how to build efficient long-context models into a precisely measurable, directly actionable form—one for which all infrastructure now exists.

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275 **A. Implementation Failure Modes**
276 **(MOHAWK+LAWCAT+LLaMA-3.1-8B)**
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Table 3. Ten previously undocumented integration failure modes and resolutions.

#	Issue	Root Cause	Resolution
1	Model ID error	meta-llama/Llama-3-8B doesn't exist	Use meta-llama/Llama-3.1-8B
2	C4 rate limit (HTTP 429)	allenai/c4 unavailable	Use monology/pile-uncopyrighted
3	AutoModelForCausalLM failure	MOHAWK uses custom SSM architecture	Use MOHAWK lazy_init mode=inference
4	LAWCAT dataset missing	c4_distill module not in LAW-CAT repo	Use alpaca_clean native loader
5	norm_epsilon attribute	LlamaBlock lacks attribute	getattr(..., 'norm_epsilon', 1e-5)
6	allow_unexpected_keys	lm_head.weight flagged	Set allow_unexpected_keys: true
7	torchrun path	Broken PATH in youra conda env	Use Path(sys.executable).parent / "torchrun"
8	Data loader chain	Missing HFDataset→Tokenize→Packing	Implemented full chain
9	Port conflict	Stale torchrun process on port 29501	Use --master_port 29502
10	Hybrid-4 architecture	MOHAWK hybrid API incompatibility	Rewrote using LayeredMambaLM